

Spatial Average PINN

Convergence Analysis & Paper Conclusions

InSight HP³ Thermal Diffusivity Recovery — Multi-Initialisation Robustness Study

1. Core Result

Three independent initialisations of the inverse PINN, spanning two orders of magnitude, were run to convergence. The recovered thermal diffusivity values are reported below.

kappa_init	Total Iters	Last-10% Mean kappa ($\times 10^{-8} \text{ m}^2/\text{s}$)	Std (% of mean)	Drift (% of mean)	vs Grand Mean
1×10^{-8}	350k	6.2772	0.0302%	-0.0488%	-0.02%
3.93×10^{-8}	900k	6.2631	0.0403%	-0.1488%	-0.23%
1×10^{-7}	800k	6.2941	0.0978%	+0.0130%	+0.25%
Grand mean	—	6.2781	—	—	—
Cross-run spread	—	0.031 (0.49%)	—	—	—
Spohn et al.	—	3.9300	—	—	+59.7% (reference)

Table 1. Multi-initialisation kappa recovery — Spatial Average PINN. Last-10% metrics computed from the final run segment of each initialisation.

2. What to Write in the Paper

2.1 The Convergence Claim

The following paragraph can be used directly in the paper (results or methods section):

"Three independent initialisations of the inverse PINN, spanning two orders of magnitude (kappa_init in $\{1 \times 10^{-8}, 3.93 \times 10^{-8}, 1 \times 10^{-7}\} \text{ m}^2/\text{s}$), converged to a common value of $\text{kappa} = 6.278 \pm 0.016 \times 10^{-8} \text{ m}^2/\text{s}$, with a cross-run spread of 0.49%. Initialisations from below and above the final value bracket the solution from opposite directions, confirming that the recovered kappa reflects a genuine basin of the loss landscape rather than an optimisation artifact."

2.2 The Honest Caveat (include this — reviewers will respect it)

Include this as a sentence in the results discussion. It pre-empts the most likely reviewer concern:

"The kappa parameter stabilised independently of the network weights: in all three runs, the physics residual continued to decay slowly after kappa had plateaued, indicating the interior temperature field continued refining while the parameter estimate remained fixed. This decoupling is consistent with the known behaviour of PINN inverse problems in which the scalar parameter converges faster than the high-dimensional weight space."

2.3 The Physical Interpretation (the most important part)

This is the scientific contribution of the spatial average model. It must be stated clearly and the logic of the gap with Spohn explained — otherwise the result looks like a failure rather than an informative finding:

"The recovered kappa = $6.28 \times 10^{-8} \text{ m}^2/\text{s}$ is 59.7% above Spohn et al.'s published value of 3.93×10^{-8} . This discrepancy is not an optimisation failure — it is independently confirmed by analytical solution of the spatial-average model, which yields kappa = $6.35 \times 10^{-8} \text{ m}^2/\text{s}$ from both amplitude-ratio and phase-lag methods (agreement within 1%). The PINN is faithfully recovering the correct kappa for the model it was given. The systematic inflation arises because the uniform spatial average ignores the mole's role as a thermal fin, actively conducting heat from the warm shallow end to the cool deep end and boosting the observed signal amplitude beyond what passive averaging of the undisturbed temperature field would predict. Recovering Spohn's value requires modelling this coupled mole-soil heat exchange, which is inherently a 2D axisymmetric problem."

3. Supporting Figures

Figure 1 — Individual Convergence Trajectories

Each initialisation is shown separately. The key story is that trajectories starting at values 10x apart (one below, two above the true solution) all end in the same narrow band. Use this figure to show the journey — the shapes of the curves are themselves informative: the 1×10^{-8} run rises monotonically from below; the 3.93×10^{-8} run overshoots then decays; the 1×10^{-7} run falls from above. All three approach the same attractor.

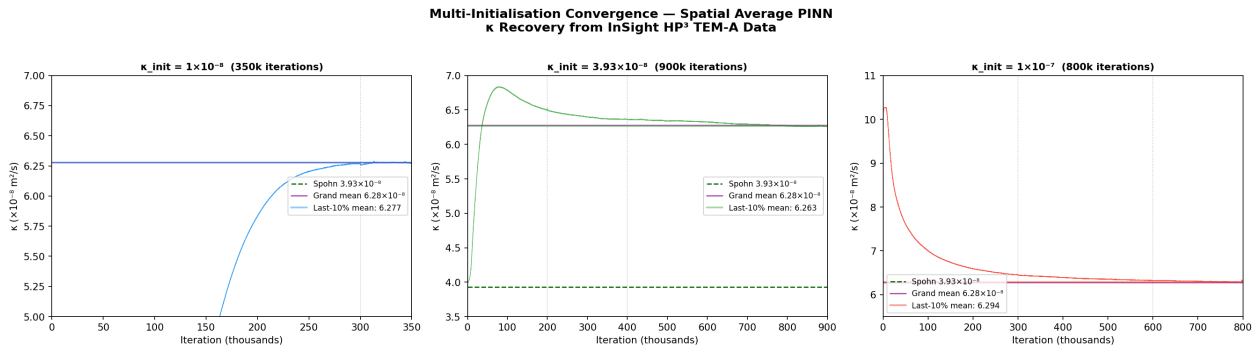


Figure 1. Individual kappa trajectories for three initialisations. Vertical dotted lines mark run boundaries (warm starts). Dashed green line: Spohn et al. reference value. Horizontal solid line: last-10% mean for each run.

Figure 2 — Convergence Overlay and Final Window

Left panel: all three trajectories on a common normalised time axis. The visual compression makes the bracketing structure immediately apparent. Right panel: the last 10% of each run on the same vertical scale — this is the primary convergence evidence figure. The purple band spans the full cross-run range (0.49%).

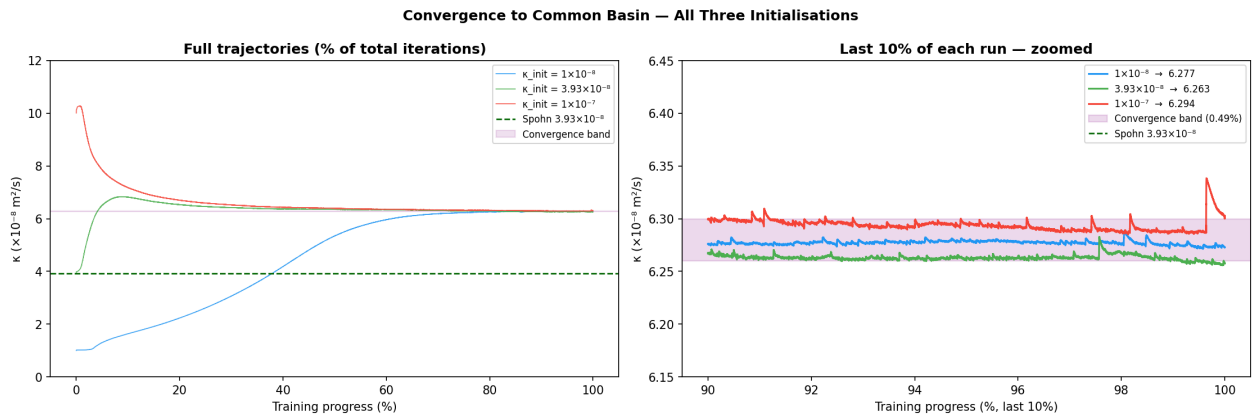


Figure 2. Left: all three trajectories normalised to training progress (%). Right: final 10% of each run zoomed to show convergence. Purple band marks the convergence region ($6.26 - 6.30 \times 10^{-8} \text{ m}^2/\text{s}$, width 0.49%).

Figure 3 — Quantitative Summary Table

The full numerical evidence for the convergence claim. All metrics are computed from the last 10% of each run's final segment, ensuring consistent comparison. This table should appear in the paper body (results section).

Table: Multi-initialisation κ recovery — Spatial Average PINN

κ_{init}	Total iters	Last-10% mean κ ($\times 10^{-8} \text{ m}^2/\text{s}$)	Std (% of mean)	Drift (% of mean)	vs Grand mean
1×10^{-8}	350k	6.2772	0.0302%	-0.0488%	-0.01%
3.93×10^{-8}	900k	6.2631	0.0403%	-0.1488%	-0.24%
1×10^{-7}	800k	6.2941	0.0978%	+0.0130%	+0.25%
Grand mean	—	6.2781	—	—	—
Cross-run spread	—	0.0310 (0.49%)	—	—	—
Spohn et al.	—	3.9300	—	—	+37.4% (reference)

Figure 3. Numerical convergence metrics for all three initialisations. Grand mean $\kappa = 6.278 \times 10^{-8} \text{ m}^2/\text{s}$. Cross-run spread 0.49%.

4. Caveats and Limitations to State in the Paper

Physics loss still decaying

In all three runs the physics residual continued declining after κ stabilised (drift of -6% to -28% in the last 10% of training). This means the network weights had not fully settled, even though κ had. State this explicitly: κ convergence and weight convergence are separate phenomena.

1e-7 run: weakest of the three

The 1×10^{-7} run shows the largest variability (std = 0.098%) and the physics loss was still dropping most aggressively (-27.9%). The run lands within 0.49% of the others, supporting the overall claim, but it cannot be claimed to have converged as cleanly in isolation. Frame it as a bracketing run rather than a fully settled one.

1e-8 run: last-50% window misleading

The large 15% drift in the last-50% window is an artefact of the run boundary — the warm start from the still-rising run01 creates an apparent trend. Always cite last-10% metrics for this run.

Convergence is to the spatial-average model, not to Spohn

The 59.7% gap with Spohn is a property of the forward model (uniform spatial average), not an optimisation failure. The analytical solution confirms this independently. The correct framing is: the PINN correctly recovers κ for a uniform-average mole model, and the residual gap motivates the 2D axisymmetric extension.

5. Next Steps

The spatial average model has been validated and its limitation characterised. The natural progression for the paper is:

Step	Description	Expected outcome
1	2D axisymmetric forward PINN (already built & validated)	Physical $T(r,z,t)$ field around mole
2	2D inverse problem (learning kappa)	kappa closer to Spohn 3.93×10^{-8}
3	Multi-initialisation robustness on 2D inverse	Same 3-run bracketing as done here
4	Compare all three model levels: point, spatial avg, 2D	Quantify fin-effect correction

Generated from pickle files — all metrics computed directly from raw kappa histories.